UEBA Module for SIEM

Project: UEBA Module for SIEM | Universidade de Aveiro

- Diogo Silva 108212
- Martim Carvalho 108749

- Dataset: X=1 (108212 + 108749 = 216961 → last digit = **1**)

Introduction

UEBA (User and Entity Behavior Analytics)

UEBA is a security approach that builds statistical profiles of normal behaviour from historical data and uses them to flag deviations in new observations. Unlike signature-based detection, which looks for known attack patterns, UEBA detects anomalies that are devices or users behaving in ways inconsistent with their own established baseline. This makes it effective against novel threats and insider attacks that leave no known signature.

GOAL

The goal of this project is to implement a UEBA module. The module reads network flow data, computes per-IP statistical baselines from a training dataset, applies a set of anomaly detection rules to a test dataset, and forwards any flagged alerts to the Wazuh manager via syslog. The full pipeline runs as a single Python script (ueba.py) that produces a consolidated report of anomalous devices and their confidence levels.

Dataset

The assigned dataset is ready to explore. It contains two distinct populations: internal clients (192.168.101.x), representing corporate workstations on the internal network, and external clients (188.83.72.x), representing users accessing a corporate server from outside. Both populations include a training file, used to establish normal behaviour, and a test file, where anomalies are detected.

▷ AI Prompt

All the context above was given to the AI exactly as is, just to understand the goal of the project.

Data Reading & First Observations

Data Loading

The first step was loading the four JSON files and understanding the structure before writing any detection logic. Each file represents one full day of network flows, with 7 fields per record: src_ip, dst_ip, port, timestamp, up_bytes, down_bytes, and protocol.

Code used:

```
int_train = pd.read_json(INTERNAL_TRAIN)
int_test  = pd.read_json(INTERNAL_TEST)
ext_train = pd.read_json(EXTERNAL_TRAIN)
```

```
ext_test  = pd.read_json(EXTERNAL_TEST)
```

Output received:

```
internal_train : (890749, 7)
internal_test  : (1008425, 7)
external_train : (712488, 7)
external_test  : (681696, 7)
```

Protocols and Ports

The first thing we explored after loading was what kind of traffic actually exists in the dataset. Grouping by port across the entire internal training file reveals something immediately striking. Concluded that only two ports exist: TCP/443 (HTTPS) and UDP/53 (DNS). No other protocols, no other ports, anywhere in the dataset. The external dataset contains only TCP/443, no DNS traffic at all, consistent with external clients communicating exclusively with the corporate public server.

Code used:

```
int_train['port'].value_counts()
# 443 (HTTPS)
# 53 (DNS)
```

Every anomaly we detect will be either in HTTPS behaviour or DNS behaviour, there is nothing else to look at.

Internal Network Topology

Inspecting the data, grouping the internal data by destination IP, three internal servers emerge consistently across all 198 clients:

Role	IP	Port
DNS Server (primary)	192.168.101.226	53
DNS Server (secondary)	192.168.101.229	53
HTTPS Server	192.168.101.240	443

Every one of the 198 internal clients communicates with all three servers during training with no exceptions. This uniformity is itself a baseline: the network behaves like a corporate environment where all machines follow the same configuration. Any device that deviates from this pattern in the test period stands out immediately.

Internal HTTPS traffic (port 443) does not all go to the same place. Grouping training flows by destination IP reveals two distinct populations: traffic to the internal HTTPS server and traffic to external HTTPS servers.

Code used:

```
https = int_train[int_train['port'] == 443]
# Identify distinct destination IPs in HTTPS traffic
dst_ips = https['dst_ip'].unique()
```

```
internal_dsts = [ip for ip in dst_ips if ip.startswith('192.168.')]
print("Internal HTTPS destinations:", internal_dsts)

" Internal HTTPS destinations: ['192.168.101.240'] "
```

The only internal destination is .240, already known from the network topology. Every other destination IP is a public address, confirming a clean two-population split. Per-client stats were then computed separately for each group:

```
INTERNAL = '192.168.101.240'
https_int = https[https['dst_ip'] == INTERNAL]
https_ext = https[https['dst_ip'] != INTERNAL]
for label, grp in [('Internal .240', https_int), ('External', https_ext)]:
  per_ip = grp.groupby('src_ip').agg(up=('up_bytes','sum'), down=('down_bytes','sum'))
  per_ip['ratio'] = per_ip['down'] / per_ip['up']
  print(f"{label}: flows={len(grp):,}  up/client={per_ip['up'].mean()/1e6:.1f} MB
  ↪  ratio={per_ip['ratio'].mean():.4f} ± {per_ip['ratio'].std():.4f}")

" Internal .240:  flows=157,055  up/client=9.0 MB   ratio=9.2255 ± 0.5809 "
" External HTTPS: flows=627,522  up/client=36.1 MB  ratio=9.2460 ± 0.2511 "
```

Both groups were characterised separately:

Destination	Flows	Up (mean/client)	Down (mean/client)	Ratio mean	Ratio std
Internal server .240	157,055	9.0 MB	83.2 MB	9.2255	0.5809
External HTTPS servers	627,522	36.1 MB	333.6 MB	9.2460	0.2511

External HTTPS servers account for 80% of all HTTPS flows. Both groups maintain nearly identical down/up ratios (∼9.23 vs ∼9.25) that means the clients download roughly 9.2 bytes for every byte they upload, a heavily download-dominant pattern typical of web browsing.

The std is computed across all 198 clients, it measures how much each individual client's ratio deviates from the group mean. The external HTTPS traffic is tighter (0.25) than traffic to .240 (0.58), confirming the two groups are statistically homogeneous on this dimension.

This confirms that the ratio is a structural property of the connection type rather than a destination-specific artefact. The combined per-client upload mean of ∼45 MB is the direct sum of the two groups (9 MB to .240 + 36 MB to external). This validates using the combined aggregate as the baseline for the HTTPS exfiltration rule, the two populations are statistically homogeneous on the ratio dimension.

DNS Invariant

The most important single observation in the entire dataset came from inspecting DNS destination IPs. During training, every DNS query from every internal client goes exclusively to .226 or .229, without a single exception across all 198 clients and 890K rows.

Code used:

```python
dns = int_train[int_train['port'] == 53]
internal_servers = set(dns['dst_ip'].unique())
# {'192.168.101.226', '192.168.101.229'}
```

Output received:

```
Internal DNS servers: {'192.168.101.229', '192.168.101.226'}
```

This is an invariant, not a tendency. It directly motivates one of the DNS detection sub-rules: any internal client that sends a DNS query to an IP outside this set during the test period is immediately anomalous.

External Clients

The external dataset contains clients from the 188.83.72.x subnet connecting to a corporate public server over HTTPS only, no DNS traffic exists in the external files. During training, these clients show a remarkably consistent down/up byte ratio across all 196 devices:

Metric	Value
Mean down/up ratio	8.50
Std	0.04
Min	8.40
Max	8.61

A standard deviation of 0.04 on a ratio of 8.50 means each client downloads almost exactly 8.5× what it uploads, consistently, across every external user. The training range spans only [8.40, 8.61]. This is the tightest baseline in the entire dataset: any test client outside this range is exhibiting behaviour no training client ever produced.

The signal is asymmetric in practice. Clients *below* the lower bound upload proportionally more to the server, suggesting unusual data being pushed toward the corporate server. Clients *above* the upper bound download slightly more (a weaker signal, since the absolute deviation is small and ratios up to 9.22 are routine for internal clients during normal browsing). Both groups are flagged, but they carry different practical weight. A 3σ window of [8.38, 8.62] is sufficient to identify all clients outside the observed training range.

The script also characterised the external inter-flow intervals during baseline computation:

Metric	Value
Mean	8.076 s
Median	1.040 s
Std	118.794 s
p90	1.910 s
p95	34.750 s

The gap between median (1.04 s) and mean (8.08 s), and the jump from p90 (1.91 s) to p95 (34.75 s), confirms a heavy-tailed distribution, consistent with human web browsing: short active bursts separated by reading pauses, with occasional long overnight idle gaps. This burst-and-pause pattern also establishes what "normal" external timing looks like, directly motivating the anti-human pattern rule (Step 2b): any external client whose inter-flow jitter is tighter than the most regular training client (16.16 s) is operating outside this observed human envelope.

▷ AI Prompt

> *Explore the data given in the dataset1 folder having in mind all the context I gave you before. The script should read the training data and output useful information that we could use in order to achieve our goal. The idea is to use a Python script, using the provided example (sampleScript.py) as a base.*

Baseline Computation

Before any detection logic runs, compute_baselines() reads both training files and builds a dictionary of statistical profiles that every rule will consume. This function computes per-IP aggregates (total upload, DNS flow counts, inter-flow intervals, destination countries) and derives the thresholds that define "normal" for each detection rule. Nothing is flagged here. The goal is purely to characterise the training period so the test period can be compared against it. The function follows a single design principle: compute everything once, return it all as a dictionary. The naive alternative would be to re-read and re-aggregate the training data inside each rule function.

This means scanning ~900k rows five separate times and duplicating the same groupby operations. Instead, compute_baselines() runs exactly once at startup, builds every statistic every rule will ever need, and passes the result as a single dict b. Each rule receives b as its only input and reads from it without modifying it.

Code Implementation

```
# HTTPS: group port-443 flows per client, compute total upload per IP
  https = int_train[int_train['port'] == 443]
  https_per_ip = https.groupby('src_ip').agg(total_up=('up_bytes','sum'),
↪  total_down=('down_bytes','sum'))

# BotNet: compute per-protocol inter-flow interval std per client
  https_sorted = https_train.sort_values(['src_ip','timestamp'])
  https_sorted['interval'] = https_sorted.groupby('src_ip')['timestamp'].diff()
  https_interval_std_per_ip = https_sorted.groupby('src_ip')['interval'].std()

  dns_sorted = dns_train.sort_values(['src_ip','timestamp'])
  dns_sorted['interval'] = dns_sorted.groupby('src_ip')['timestamp'].diff()

# Geo/Destination: build global sets - union of all ASNs and internal dst_ips seen by any
↪  client in training
  global_train_asns = set(pub_train['dst_ip'].apply(get_asn).dropna())
  global_train_internal_dsts = set(priv_train['dst_ip'].unique())

# Fan-out: unique external dst_ip count per src_ip (Step 4c)
  ext_fan = pub_train.groupby('src_ip')['dst_ip'].nunique()

# New source IPs: set of all src_ips seen in training (Step 1)
  train_src_ips = set(int_train['src_ip'].unique())
```

```
# DNS: count flows per client, extract the set of destination servers
  dns_per_ip = dns.groupby('src_ip').agg(dns_flows=('up_bytes','count'))
  dns_internal_servers = set(dns['dst_ip'].unique())
```

The dictionary compute_baselines() returns one key per metric below. Each rule function receives this dictionary as its only input and reads the relevant keys no rule recomputes or re-reads training data on its own.

Threshold table

All thresholds are derived dynamically from the training data.

```
New source IP:                 none - binary set membership check
HTTPS upload threshold:        116.6 MB          (mean+3σ)
DNS flow threshold:            1399 flows        (mean+3σ)
BotNet HTTPS interval p05:     18.81 s
BotNet DNS  interval p05:      63.73 s
External ratio window:         [8.3817, 8.6226]
Geo min intensity flows:       10 flows
External fan-out threshold:    258 unique dst IPs  (mean+3σ)
```

Each threshold targets a different dimension of behaviour. Not every rule requires a statistical threshold: when a training invariant holds absolutely, any test-period violation is unconditionally anomalous.

- **New source IP (Step 1)** requires no threshold at all. Any internal device that generates traffic in the test period but had zero presence during training is flagged unconditionally.
- **HTTPS upload threshold (116.6 MB)** defines the maximum total data a normal internal client sends over HTTPS in a day; anything above this suggests bulk data exfiltration.
- **DNS flow count threshold (1,399 flows)** flags clients generating an abnormal number of DNS queries, the primary signal for DNS-based C&C beaconing.
- **BotNet interval p05 thresholds** are split by protocol: 18.81 s for HTTPS and 63.73 s for DNS. Any client below its protocol's floor is more regular than 95% of all legitimate training clients. Separating the protocols prevents a device with naturally irregular HTTPS activity from masking a tight DNS beaconing pattern.
- **External ratio window ([8.3817, 8.6226])** defines the expected down/up range for external clients. Any external client outside this window is interacting with the corporate server in an anomalous way.
- **Geo minimum intensity floor (10 flows)** filters CDN noise: any client reaching a country new to the entire network with fewer than 10 flows is treated as a CDN edge rotation rather than a deliberate new destination.
- **External fan-out threshold (258 unique IPs)** flags any internal client contacting an anomalous number of distinct external destinations in a single day. Derived from mean (118.1) + 3σ (3×46.7), it catches sweep-style behaviour where a host checks in with many scattered external endpoints.

The 3σ Rule

The choice of 3σ as the threshold is an industry standard in statistical anomaly detection. Under a normal distribution, 99.9% of observations fall within 3 standard deviations of the mean, meaning only 0.1% of legitimate traffic would be flagged as anomalous by chance.

Applied consistently across all volume-based rules, it means the thresholds adapt automatically to the dataset: a network with higher baseline DNS traffic will produce a higher DNS threshold, and a network with tighter HTTPS upload patterns will produce a lower upload threshold. No manual tuning required.

The alternatives were evaluated and rejected:

- **2**σ (97.7% coverage) would produce approximately 4 false positives per rule per day, creating a noise floor that drowns real signals.
- **4**σ (99.9937% coverage) would set the bar so high that moderate exfiltration (a device uploading 200 MB when the mean is 45 MB) would go undetected.

Country Distribution

As part of the baseline computation, every internal client's HTTPS traffic was mapped to destination countries using a GeoIP database. Across all 198 clients and the full training period, the network contacted 36 unique countries. The top 5 alone account for 94.8% of all flows, showing a heavily concentrated geographic footprint:

Country	Flows	% of Total
PT	234442	37.4%
US	164761	26.3%
CA	93719	14.9%
FR	64307	10.2%
NL	37617	6.0%
(others)	32676	5.2%

These 36 countries define the known geographic footprint of this network, any country contacted in the test period that never appeared in training is an immediate network-level anomaly, regardless of which client triggered it.

▷ AI Prompt

Now we need to implement a `compute_baselines()` function to define the baselines. The idea is to compute the necessary statistics and thresholds that will be used by the detection rules. Use the information gathered during data exploration and the list of rules to decide which statistics are relevant for each rule. The goal is to find the "normal" thresholds from the training data, so that when we apply the rules to the test data, any deviations from these baselines can be flagged as anomalies.

Rule Implementation

With the baselines established, the next step was implementing the detection rules. Each rule targets a specific type of anomalous behaviour and operates independently, consuming the baseline dictionary computed in the previous step. Each rule follows the same pipeline: statistical thresholds are derived exclusively from the training dataset and then applied to the test dataset, which represents a separate day where anomalous behaviour may be present. The training data never sees the test data, and the test data never influences the thresholds.

This strict separation is what gives the detection statistical validity: the baselines describe what normal looks like before any anomaly occurs, and the rules measure how far the test behaviour deviates from that established normal. The rules were designed to be complementary rather than redundant: cumulative volume rules catch both bulk attacks and low-and-slow exfiltration that rate-based detection would miss, while timing and behavioural rules catch threats that leave no volumetric footprint at all. For each rule, the process followed the same structure: define the metric, compute the threshold from training, apply it to the test data, and examine what gets flagged.

In several cases the first approach produced too many false positives or broke down statistically, requiring iteration before arriving at a clean result.

Rule Design Rationale

The rules are designed around the project's five detection requirements. The table below maps each requirement to the implemented step(s) and the signal dimension each measures.

Spec Requirement	Step(s)	Signal dimension
(i) BotNet activities	Step 6	Inter-flow timing regularity per protocol
(ii) HTTPS data exfiltration	Step 3	Upload volume
(ii) DNS data exfiltration	Step 5 sub-rule 1	DNS query volume
(iii) C&C activities using DNS	Step 5 sub-rule 2 + Step 6 (DNS)	Resolver bypass + timing regularity
(iv) Anomalous external destinations	Steps 4, 4b, 4c	Geography, ASN infrastructure, unique IP count
(v) External users anomalous	Steps 2, 2b	Down/up byte ratio + inter-flow timing regularity
(bonus) Rogue/new devices	Step 1	Set membership — no threshold needed

Step 1 — New Source IP Detection

What to look for

The most fundamental principle in UEBA is that a behavioral baseline can only describe devices that were actually observed during training. Every statistical rule in this pipeline produces a number that is compared against a distribution computed from training data. But this comparison is only meaningful if the device itself has a training history. A device with zero training presence has no baseline to deviate from, not a small deviation, but a complete absence of normal.

In a corporate environment, every legitimate device should appear in the training window. A new device appearing in the test period is inherently suspicious: it could be a rogue endpoint physically connected to the network, a new implant deployed by an attacker after the training window, or a compromised host that changed its IP to evade rules tuned on the original identity. Any of these scenarios warrants immediate investigation regardless of what the device is actually doing in terms of volume or timing.

This rule requires no statistics and no threshold. The question is binary: was this source IP seen at all during training?

Code Implementation

```
train_ips = baselines['train_src_ips']                    # set of all training src_ips
new_ips   = set(int_test['src_ip'].unique()) - train_ips
```

One set subtraction. A device is either in the training set or it is not.

Results

The rule flagged 2 internal IPs:

```
[ALERT] 192.168.101.11
  Why: 1508 flows (1309 HTTPS, 199 DNS), HTTPS upload: 14.3 MB,
       unique ext IPs: 64 - all metrics sub-threshold,
       absent entire training period
[ALERT] 192.168.101.93
  Why: 3788 flows (3286 HTTPS, 502 DNS), HTTPS upload: 38.0 MB,
       unique ext IPs: 101 - all metrics sub-threshold,
       absent entire training period
```

Both devices generate thousands of flows in the test period yet were completely invisible to every other rule in the pipeline. The reason is instructive: their traffic profiles are deliberately sub-threshold. .11 uploads only 14.3 MB over HTTPS (threshold: 116.6 MB), contacts only 64 unique external IPs (threshold: 258), and generates only 199 DNS queries (threshold: 1,399). .93 uploads 38 MB, contacts 101 unique external IPs, and generates 502 DNS queries. Every volume metric sits well below every threshold. This is the profile of a stealthy implant: generate enough traffic to operate, but stay below every detection limit on every dimension. The only rule that catches them is the one that requires no threshold at all.

A complementary observation: the training set contains 2 IPs that disappeared entirely in the test period: The 192.168.101.57 (2,117 training flows) and 192.168.101.202 (2,501 training flows). While device turnover is normal and these are not flagged as alerts, the simultaneous disappearance of two training devices and appearance of two new ones in the same test window is a pattern consistent with IP reassignment following lateral movement. It is noted here as investigative context for an analyst reviewing the full picture.

Spec requirement covered: bonus rule beyond the minimum.

Step 2 — External Anomaly Detection

What to look for

The external anomaly rule targets clients from the 188.83.72.x subnet accessing the corporate server in an unusual way. The key insight here is that the relevant signal is not the amount of traffic, but the ratio between downloaded and uploaded bytes. During training, every external client consistently downloads roughly 8.5× what it uploads: a tight, stable pattern with a standard deviation of just 0.04.

Note that this 8.50 baseline is distinct from the 9.22 ratio observed for internal clients in the HTTPS destination split analysis. The two ratios measure opposite directions of traffic: 9.22 describes internal clients downloading from external internet servers (web browsing), while 8.50 describes external clients downloading from the corporate server (server access). Different populations, different servers and different baselines.

This makes the ratio the most discriminating metric in the entire dataset: a client whose ratio breaks this pattern is interacting with the server in a fundamentally different way than normal, regardless of total volume. A 3σ window of [8.3817, 8.6226] is sufficient to flag real anomalies with minimal false positives.

Code Implementation

Code used:

```
low = mean - SIGMA * std  # 8.3817
high = mean + SIGMA * std # 8.6226

flagged = per_ip[(per_ip['ratio'] < low) | (per_ip['ratio'] > high)]
```

We compute the ratio for each external client in the test period and flag anyone outside the 3σ window. Below the lower bound means the client is uploading proportionally more than expected. Above the upper bound means it is downloading proportionally more than expected.

Results

The rule flagged 5 external IPs, which naturally split into two distinct behavioural groups:

```
[ALERT] 188.83.72.61   ratio=8.232  deviation=6.7σ
[ALERT] 188.83.72.64   ratio=8.338  deviation=4.1σ
[ALERT] 188.83.72.174  ratio=8.334  deviation=4.2σ
[ALERT] 188.83.72.182  ratio=8.661  deviation=4.0σ
[ALERT] 188.83.72.210  ratio=8.644  deviation=3.5σ
```

The results revealed two groups:

- The first group (.61, .64, .174) sit below the lower bound. Their ratio is lower than normal, meaning they upload proportionally more than a legitimate client would. This is consistent with a compromised account being used to push data toward the corporate server, or with reversed exfiltration where data is staged on the server from outside.
- The second group (.182, .210) sit above the upper bound. They are flagged because the rule is symmetric and they cross the statistical threshold, but the practical deviation is minimal: ratios of 8.64–8.66 against a baseline of 8.50 represent an absolute difference of just 0.14–0.16. Because the threshold is already extremely tight (std = 0.04), any IP above the upper bound is by construction close to it. In contrast, internal clients routinely browse at ratios of 9.22 without any concern. These two IPs are retained as MEDIUM confidence anomalies because they do break a tight statistical invariant, but analysts should treat them as the weakest signals in the pipeline and prioritise the upload-heavy group.

The 188.83.72.61 at 6.7σ is the most extreme anomaly in the entire external dataset.

Spec requirement covered: (v) anomalous behaviour of external clients, via down/up byte ratio deviation.

Step 2b — External Automation (Anti-Human Pattern)

What to look for

Human browsing behaviour is inherently irregular: users pause, read, and navigate at unpredictable intervals. A client whose inter-flow timestamps are unnaturally consistent (more regular than any client ever observed in training) is exhibiting a pattern incompatible with human operation.

The detection metric is the per-IP standard deviation of inter-flow intervals (*jitter*) in the test period. Jitter measures how much the gaps between consecutive flows vary: high jitter means the client connects at unpredictable times (human-like); low jitter means it connects on a fixed clock with almost no drift (machine-like). Across all 196 external training clients, the most regular client had a jitter of 16.16 s. This value is the natural floor of human-like behaviour in this dataset.

It is important to note that machine-like regularity is not inherently malicious. Legitimate automated consumers (like monitoring probes, scheduled API integrations, or health-check services) also connect at fixed intervals by design. The regularity pattern is a strong indicator that warrants analyst attention, but it cannot distinguish between a bot attacking the server, malware checking into an internal resource, or a benign automated service.

Code Implementation

```
# Threshold: minimum per-IP interval std across all 196 training clients
# Interpretation: no training client was ever more regular than this
ext_iv_std_per_ip = ext_sorted.groupby('src_ip')['timestamp']
                    .apply(lambda x: x.diff().std()).dropna()
threshold = ext_iv_std_per_ip.min()  # 16.16 s

flagged = iv_std_test[iv_std_test < threshold]
```

Results

The rule flagged 5 external IPs, all far below the 16.16 s training floor:

```
[ALERT] 188.83.72.185  std=0.51s  median=0.91s  (32x below training floor)
[ALERT] 188.83.72.32   std=2.02s  median=8.00s  ( 8x below training floor)
[ALERT] 188.83.72.46   std=2.02s  median=8.00s  ( 8x below training floor)
[ALERT] 188.83.72.48   std=2.03s  median=8.00s  ( 8x below training floor)
[ALERT] 188.83.72.166  std=2.00s  median=8.00s  ( 8x below training floor)
```

Four of the five (.32, .46, .48, .166) share near-identical statistics: approximately 2 s standard deviation and an 8-second median interval. This uniformity strongly suggests a shared automated tool or configuration connecting to the corporate server on a fixed schedule. The fifth (.185) is even more extreme at 0.51 s standard deviation with sub-second timing precision, categorically incompatible with human interaction.

A natural concern is whether a 2 s standard deviation is genuinely abnormal or simply a tight but plausible human pattern. The test data resolves this decisively: ranking all 196 external test IPs by their interval standard deviation reveals a clean bimodal split with no IP between 2.03 s and 18.39 s. There is an 8-second empty gap separating the two clusters. If 2 s were a natural point on a continuum, IPs at 4 s, 8 s, or 12 s would exist. That combination of fixed median interval and near-zero jitter (2 s) is the signature of a configured timer, not a human.

Whether these represent malicious bots, coordinated scanners, or legitimate monitoring agents cannot be determined from timing alone. They are flagged as **MEDIUM** confidence anomalies requiring analyst follow-up.

Spec requirement covered: (v) anomalous behaviour of external clients: inter-flow jitter (anti-human pattern).

Step 3 — Internal Anomaly Detection (HTTPS Exfiltration)

What to look for

The HTTPS exfiltration rule targets internal clients sending an abnormal amount of data outbound over HTTPS. The detection metric is total upload volume per client: any client uploading more than mean + 3σ (116.6 MB) in the test period is flagged. Measuring cumulative volume over the full observation window, rather than peak rate or per-session bytes, is a deliberate design choice: a low-and-slow exfiltrator operating at a modest rate stays invisible to rate-based rules, but its data accumulates across the test period and will eventually cross the total volume threshold.

A second signal, the PCR (Producer-Consumer Ratio), was implemented and evaluated before finalising this rule. PCR = (up − down) / (up + down) measures the upload/download balance independently of volume. The training mean was –0.8046 (heavily download-dominant), and the 3σ threshold was –0.7918. Applied to the test data, PCR flagged four additional IPs with upload volumes well below the training mean. On closer inspection, these were false positives: with few HTTPS flows, natural variance in the upload/download balance is enough to push PCR across the threshold. PCR was removed; the devices it would have caught are independently and more reliably detected by the BotNet beaconing rule through timing analysis.

Code Implementation

```
flagged = per_ip['total_up'] > vol_threshold  # > 116.6 MB
```

The threshold is derived from combined HTTPS traffic (internal .240 + external servers). Both groups show near-identical down/up ratios (∼9.23), confirming the combined aggregate is a valid baseline for volume normalisation.

One metric was evaluated and removed before arriving at this design: the raw down/up ratio (total_down / total_up), which is the same metric used for external clients. An equivalent internal ratio was computed and applied to the test data. It flagged the identical set of IPs as the volume threshold: zero additional detections. The raw ratio approach was dropped in favour of the simpler volume threshold alone.

Results

The rule flagged 6 internal IPs:

```
[ALERT] 192.168.101.187  upload=7586 MB  deviation=316σ
[ALERT] 192.168.101.14   upload=5352 MB  deviation=223σ
[ALERT] 192.168.101.208  upload=4402 MB  deviation=183σ
[ALERT] 192.168.101.26   upload=259 MB   deviation=9σ
[ALERT] 192.168.101.197  upload=138 MB   deviation=4σ
[ALERT] 192.168.101.207  upload=119 MB   deviation=3σ
```

All six devices exceed the 116.6 MB threshold cleanly. The top three (.187, .14, .208) represent bulk data dumps at 4.4–7.6 GB, 183–316σ above baseline, that are orders of magnitude beyond any legitimate client in training. The remaining three (.26, .197, .207) sit between 119 MB and 259 MB, each clearly distinguishable from the highest unflagged device at 105.2 MB.

Spec requirement covered: (ii) HTTPS data exfiltration via upload volume anomaly.

Step 4 — GeoDestination Anomaly Detection

What to look for

The geo destination rule targets internal clients contacting countries that are new to the entire network. The first approach was straightforward: flag any internal client that contacts a country in the test period that it never contacted during training.

The result was immediate and catastrophic: 163 of 198 clients flagged, an 82% false positive rate. The reason is CDN rotation: large content delivery networks distribute their infrastructure globally and rotate IP addresses across countries regularly, meaning a legitimate user visiting the same website on two different days may hit servers in different countries each time. A naive per-IP new-country rule is completely blind to this and produces noise, not signal.

Code Implementation

To solve the false positive problem, the naive per-IP approach was replaced with a global detection strategy: flag any client reaching a country that no client in the entire network contacted during training. Any access to such a country is immediately anomalous at the network level, it means not just new to the individual deviced but new to the whole organisation.

```
new_to_network = test_countries - global_train_countries

if new_to_network and flows_to_new >= MIN_INTENSITY_FLOWS:
      # fire alert
```

The set difference operation is the core: subtracting the network-wide known country set from the observed destinations leaves only genuinely new territory. A minimum intensity floor removes CDN edge noise: any new-country access with too few flows is treated as a stray CDN rotation rather than a deliberate connection.

Two guards are applied in combination to eliminate CDN noise:

- **Absolute floor (3 flows):** derived from training data 10%of per-(IP, country) flow counts across 2,354 training pairs. CDN rotation artifacts typically produce 1–2 flows, well below this floor.
- **Proportion guard (1% of total traffic):** new-country flows must represent at least 1% of that IP's total test-period flows. A client with thousands of flows contacting a new

country in just 4–9 flows (0.06–0.37% of traffic) is exhibiting CDN behaviour, not deliberate communication.

Both conditions must be satisfied simultaneously. This dual guard prevents both low-volume CDN hits and high-volume IPs where a tiny fraction of traffic incidentally reaches an unknown country.

ASN (Autonomous System Number) detection was also implemented as a complementary rule: a device contacting a new cloud provider or hosting company is suspicious even if the destination country is already in the training set. ASN lookups use a separate GeoIP database and compare against the global union of all ASNs seen by any client during training. This is implemented as a separate step (Step 4b) rather than merged into this rule, since it catches a different threat model.

Results

The dual-guard rule produced 3 flagged IPs, all high-confidence:

```
[ALERT] 192.168.101.125   523 flows to new country/ies:
          BG, CZ, DK, FI, IR, LV, PL, PY, RU, SC, UA  (11.7% of traffic)
[ALERT] 192.168.101.36    904 flows to new country/ies:
          AR, BA, BG, BY, CZ, EE, FI, GE, HU, IQ, IR, KZ, LU, LV, PL, RU, UA  (13.5%)
[ALERT] 192.168.101.72   1157 flows to new country/ies:
          BD, BY, BZ, EE, FI, GR, HR, IR, KZ, LU, LV, NG, PL, RU, UA  (13.1%)
```

All three reach multiple countries new to the entire network, including Russia, Iran, and Ukraine, geographies absent from all 198 training clients. Each sends over 11% of its total test traffic to these new destinations, ruling out CDN rotation. All three are independently confirmed by Step 4b through ASN analysis, promoting them to HIGH confidence.

Two IPs (.167 and .175) had 4–9 flows to Belgium (never seen in training) but represented only 0.06–0.37% of their total traffic. These were correctly excluded as likely CDN edge rotation rather than deliberate connections.

Spec requirement covered: (iv) anomalous external destinations via new-country geography.

Step 4b — New Destination Detection

What to look for

Beyond new geographic territory, two complementary destination-based signals catch compromised devices reaching infrastructure that was simply never contacted during training regardless of country.

- Sub-rule A (New External ASN): flags any internal client contacting an Autonomous System that no client in the entire network used during training. Country detection can miss this if the new infrastructure happens to be in a country the network already visited. Comparing against the global union of all ASNs seen across all 198 training clients makes the signal noise-resistant: legitimate CDN providers used by even a single training client are whitelisted, so only genuinely unseen infrastructure triggers an alert.
- Sub-rule B (New Internal Destination): flags any internal client communicating with an internal IP (192.168.101.x) that was never a destination in any training flow. During training, all internal HTTPS traffic goes exclusively to the single known server .240.

Code Implementation

```
# Sub-rule A - new external ASN (global training set)
pub_test['asn'] = pub_test['dst_ip'].apply(get_asn)
new_asns = set(grp['asn'].dropna().unique()) - global_train_asns
if new_asns and flows >= MIN_NEW_EXTERNAL_FLOWS:  # MIN = 10
    # fire alert

# Sub-rule B - new internal destination (global training set)
new_dsts = set(grp['dst_ip'].unique()) - global_train_internal_dsts
if new_dsts and flows >= MIN_NEW_INTERNAL_FLOWS:  # MIN = 5
    # fire alert
```

The global baseline comparison is the critical design choice. Using per-client known-ASN sets instead would cause false positive explosions: CDN providers rotate IP addresses across hundreds of ASNs, so nearly every device appears to reach "new" ASNs compared to its own individual training history. The global union approach is noise-immune because any ASN touched by any of the 198 training clients is whitelisted.

Results

Step 4b flagged 6 alerts across 6 IPs:

```
[ALERT] 192.168.101.36    1015 flows to 287 new-to-network ASNs
          (incl. Russian hosting: REG.RU, C.S.T Ltd, Eneva Ltd...)
[ALERT] 192.168.101.72    1319 flows to 354 new-to-network ASNs
          (incl. Cloud Technologies LLC/Cloud.ru, AKT LLC, NPO Unimach LLC...)
[ALERT] 192.168.101.125    598 flows to 169 new-to-network ASNs
          (incl. OBIT Ltd, AS43668 LLC, REG.RU...)
[ALERT] 192.168.101.68     273 flows to new internal IPs:
          192.168.101.138, 192.168.101.186
[ALERT] 192.168.101.138    209 flows to new internal IPs:
          192.168.101.186, 192.168.101.68
[ALERT] 192.168.101.186    429 flows to new internal IPs:
          192.168.101.138, 192.168.101.68
```

The ASN detections for .36, .72, and .125 independently confirm the geo anomalies already identified in Step 4: the same three devices reach both new countries and new infrastructure, with the ASN names pointing directly at Russian hosting providers not present anywhere in training. This cross-rule corroboration promotes all three to HIGH confidence.

The lateral movement triangle (.68, .138, .186) is the most significant finding in this rule. These three internal devices communicate with each other over TCP:443, a connection that, across all 198 clients and 890 K training rows, goes exclusively to the known server .240. The fact that all three devices appear both as sources and destinations, forming a closed triangle of mutual communication, rules out a simple misconfiguration. The traffic pattern is symmetric (up/down byte ratios near 1:1, unlike the 9:1 ratio of normal web browsing), consistent with peer-to-peer command relay or lateral file transfer between compromised hosts, worth being mentioned.

Spec requirement covered: (iv) anomalous external destinations via new ASN infrastructure and new internal destinations (lateral movement).

Step 4c — External Destination Fan-out

What to look for

The fan-out rule addresses a different dimension of suspicious external behaviour: not what destinations a client reaches, but how many. During training, internal clients contact a mean of 118.1 unique external IPs per day, with a standard deviation of 46.7. A device that suddenly reaches hundreds more unique external endpoints than any legitimate client did in training is not browsing normally.

The key distinction from Step 4 and Step 4b is the signal being measured. Step 4 flags new geography; Step 4b flags new ASNs (new infrastructure providers). Step 4c is purely statistical: it fires when the raw count of unique external destinations is anomalous, regardless of whether those destinations are known or new. A device could contact only known ASNs in known countries and still fire this rule if it contacts far more of them than any training client ever did.

Code Implementation

```
ext_fan =
↪  int_test[~int_test['dst_ip'].apply(is_private)].groupby('src_ip')['dst_ip'].nunique()
threshold = ext_fan_mean + SIGMA * ext_fan_std  # 258 unique IPs

flagged = ext_fan[ext_fan > threshold]
```

The implementation is deliberately simple: one groupby, one nunique, one threshold comparison. The baseline is the per-client unique external IP count from training (mean=118.1, std=46.7), derived from the same public-IP subset already used by the geo and ASN rules.

Results

The rule flagged 4 internal IPs:

```
[ALERT] 192.168.101.72    811 unique external IPs
          (14.9σ above mean 118; threshold 258)
[ALERT] 192.168.101.36    639 unique external IPs
          (11.2σ above mean 118; threshold 258)
[ALERT] 192.168.101.125   409 unique external IPs
          (6.2σ above mean 118; threshold 258)
[ALERT] 192.168.101.207   273 unique external IPs
          (3.3σ above mean 118; threshold 258)
```

All four are already flagged by other rules, and that is precisely the point: fan-out provides a third independent confirmation for .36, .72, and .125, and a second independent axis of evidence for .207 alongside its exfiltration and DNS signals. The deviations are substantial reinforcing that these devices are not exhibiting borderline behaviour but a fundamentally different operational mode. The highest unflagged device is .114 at 234 unique IPs, sitting 24 IPs below the threshold with no other anomalous signals, confirming the threshold is not set too aggressively.

Spec requirement covered: (iv) anomalous external destinations via unique destination count (breadth of reach).

Step 5 — DNS Anomaly Detection

What to look for

The DNS anomaly rule targets internal clients exhibiting abnormal DNS behaviour. The active sub-rule is DNS-1 (Volume): it flags any client whose total DNS flow count exceeds mean + 3σ (1,399 flows), the primary signal for both DNS-based C&C beaconing and DNS tunneling.

The DNS-1 implementation:

```
# DNS-1: volume anomaly
  threshold      = mean + SIGMA * std          # 1,399 flows
  volume_flagged = dns_count[dns_count > threshold]
```

C&C Beaconing vs DNS Exfiltration

Before presenting the results, it is important to distinguish between two different DNS-based attack patterns, since they look different in the data and have different implications.

- DNS data exfiltration encodes stolen data inside DNS query subdomains: the queries are bursty, aperiodic, and contain long high-entropy subdomain labels.
- DNS C&C beaconing uses DNS queries as a heartbeat mechanism: a malware implant checks in with its C&C server at a fixed clock-driven interval, producing a very regular, high-volume query pattern with normal-looking short domain names.

Aspect	C&C Beaconing	DNS Exfiltration
Pattern	Fixed periodic intervals	Bursty, aperiodic
Query content	Short, normal-looking	Long high-entropy subdomains
Volume	Extreme query count	Moderate count, large payload
Interval	Exact, clock-driven	Irregular

What we detected in this dataset is C&C beaconing, not exfiltration: the exact 0.05s median interval across all flagged devices is the signature of a malware timer. Actual data exfiltration in this dataset happens over HTTPS, as seen in the previous rule.

A second approach was also attempted before arriving at the final: flagging anomalous DNS behaviour by per-query payload size (up_bytes). The hypothesis was that DNS exfiltration encodes data inside subdomain labels, making each query packet larger than a normal lookup. The training baseline for DNS query size is tight, applied to the test data, not a single IP exceeded this threshold. The beaconing pattern in this dataset uses fixed-size queries carrying only a short beacon identifier, not encoded data, so the payload size is indistinguishable from normal traffic. This sub-rule was computed, tested, and dropped, it has zero discrimination power for this dataset.

A third sub-rule (DNS-2, Public server) was also tested: flagging any client that sent DNS queries to a server outside the known internal resolvers (.226 and .229). Applied to the test data, DNS-2 produced zero alerts and added no new IPs to the detection list. All clients exclusively used the internal resolvers throughout the test period, so the sub-rule was removed from the final pipeline.

Results

DNS-1 flagged 5 internal IPs:

```
[ALERT] 192.168.101.41   flows=39,493  median interval=0.05s  increase=129×
[ALERT] 192.168.101.23   flows=8,651   median interval=0.05s  increase=8.6×
[ALERT] 192.168.101.201  flows=2,941   median interval=0.05s  increase=15.7×
[ALERT] 192.168.101.148  flows=1,661   median interval=0.05s  increase=4.8×
[ALERT] 192.168.101.207  flows=1,418   median interval=0.05s  increase=3.4×
```

Every flagged device shows an exact 0.05s median interval between DNS queries: a clock-driven malware timer firing consistently throughout the day. The .41 is the most extreme case at 39,493 flows, 129× its own training baseline, with queries concentrated almost entirely on the two internal resolvers.

Spec requirement covered: (ii) DNS data exfiltration (sub-rule 1: volume) + (iii) C&C activities using DNS (sub-rule 2: resolver bypass).

Step 6 — BotNet Beacon Detection

What to look for

The BotNet beaconing rule targets internal clients whose network traffic follows an unusually regular timing pattern. A botnet implant checks in with its C&C server at a fixed interval, producing a very low standard deviation in inter-flow timestamps compared to legitimate users, whose traffic is naturally irregular.

The detection metric is the standard deviation of inter-flow intervals per client, computed separately for HTTPS traffic and DNS traffic. Separating the protocols is essential: a device with irregular HTTPS browsing and regular DNS beaconing would have its DNS signal masked by the HTTPS noise in a combined calculation.

Mean – 3σ cannot be applied: both per-protocol interval std distributions are right-skewed, so the formula produces a negative lower bound for HTTPS and is equally inapplicable for DNS. No meaningful floor can be set this way on a right-skewed distribution.

Code Implementation

The threshold for each protocol is its 5% (p05) of per-client interval standard deviations from training. This is the empirical lower boundary of normal behaviour: any client in the test period with an interval std below the HTTPS floor (18.81 s) is more regular than 95% of all legitimate HTTPS sessions in training; similarly for DNS (63.73 s).

For HTTPS beaconing, a regularisation ratio check is added on top of the threshold: the device's own training-period HTTPS interval std must be at least 1.5× higher than its test-period std. This guards against devices that were naturally tight in training from being falsely flagged. A device with train_std = 20.00 s and test_std = 18.00 s has barely changed; a device with train_std = 120.00 s and test_std = 18.00 s has genuinely become more regular.

DNS beaconing includes an additional volume guard: the DNS flow count must also exceed the DNS volume threshold (1,399 flows). A device with few but regularly-timed DNS queries is not generating enough traffic to constitute meaningful beaconing.

An alternative approach, the Coefficient of Variation (CV = std/mean, with CV ≤ 0.2), was evaluated and rejected: the CV ranges for normal clients (3.93–25.52) and confirmed beacons (3.7–16.6) overlap almost completely in training, providing no clean separation point.

```
# HTTPS beaconing: below p05 AND regularisation ratio ≥ 1.5×
https_std < https_p05  and  (train_std / https_std) >= 1.5

# DNS beaconing: below p05 AND volume anomaly
dns_std < dns_p05  and  dns_count > dns_vol_threshold
```

Results

The rule flagged 6 internal IPs, split into two per-protocol beaconing channels:

```
[ALERT] 192.168.101.23   DNS  interval std=9.13s   (threshold 63.73s)  median=0.05s
[ALERT] 192.168.101.41   DNS  interval std=12.60s  (threshold 63.73s)  median=0.05s
[ALERT] 192.168.101.201  DNS  interval std=36.62s  (threshold 63.73s)  median=0.05s
[ALERT] 192.168.101.117  HTTPS interval std=10.35s (threshold 18.81s)
          median=1.00s  regularisation=8.46×
[ALERT] 192.168.101.157  HTTPS interval std=18.64s (threshold 18.81s)
          median=1.03s  regularisation=6.49×
[ALERT] 192.168.101.188  HTTPS interval std=18.27s (threshold 18.81s)
          median=1.03s  regularisation=1.84×
```

The 6 devices split cleanly by protocol:

- The DNS beacons (.23, .41, .201) fire at an exact 0.05 s median interval through the internal DNS resolvers, consistent with Step 5's findings. The same malware heartbeat confirmed by both volume and timing analysis.
- The HTTPS beacons (.117, .157, .188) fire at intervals clustering tightly between 1.00 and 1.03 seconds, a fingerprint of a single malware family sharing the same hardcoded C2 timer.

The regularisation ratio validates that each HTTPS beacon genuinely changed behaviour in the test period. .117 shows the most dramatic change: its training HTTPS interval std was ∼87.40 s (irregular browsing); in the test period it dropped to 10.35 s (8.46× more regular). .157 shows a 6.49× change. Even the borderline .188 shows a 1.84× change, comfortably above the 1.5× threshold.

Two previously considered devices were correctly excluded by the per-protocol approach. .32 and .160 appeared suspicious under a combined-channel analysis because mixing their DNS 0.05 s intervals with HTTPS ∼1.00 s intervals produced an artificially low overall std. With per-protocol analysis, .32's DNS flow count falls below the 1,399-flow volume guard, and .160's HTTPS regularisation ratio is only 1.17× (barely changed from training).

Spec requirement covered: (i) BotNet activities + (iii) C&C activities using DNS via inter-flow timing regularity per protocol.

Final Results

Final output

Running the full detection pipeline against the test dataset produced 31 unique anomalous IPs: 21 internal clients (192.168.101.x) and 10 external clients (188.83.72.x). Each flagged IP was assigned a confidence level based on how many independent rules triggered it: HIGH when two or more rules agree on the same device, MEDIUM when only one rule fires.

The two-level scheme is not arbitrary:

- When a single rule flags a device, there is always a small residual probability of a false positive.
- When two completely independent detection methods both flag the same IP on the same day, the probability of a coincidental double false positive is the product of the individual rates, dropping to approximately 0.0001% per device.

The 7 HIGH confidence IPs are therefore treated as confirmed compromised devices. The 24 MEDIUM confidence IPs are genuine anomalies that warrant investigation but cannot be confirmed by corroboration alone.

```
Total Unique Anomalous IPs: 31
   - Internal (192.168.101.x): 21
   - External (188.83.72.x): 10

 -- HIGH confidence (2+ independent rules) --
 [HIGH  ] 192.168.101.23       DNS Anomalies | BotNet Beaconing
 [HIGH  ] 192.168.101.36
            New Country Destinations | New Destination IPs / ASNs
            | External Destination Fan-out
 [HIGH  ] 192.168.101.41       DNS Anomalies | BotNet Beaconing
 [HIGH  ] 192.168.101.72
            New Country Destinations | New Destination IPs / ASNs
            | External Destination Fan-out
 [HIGH  ] 192.168.101.125
            New Country Destinations | New Destination IPs / ASNs
            | External Destination Fan-out
 [HIGH  ] 192.168.101.201      DNS Anomalies | BotNet Beaconing
 [HIGH  ] 192.168.101.207
            HTTPS Data Exfiltration | External Destination Fan-out
            | DNS Anomalies

 -- MEDIUM confidence (single rule) --
 [MEDIUM] 188.83.72.32         External Automation
 [MEDIUM] 188.83.72.46         External Automation
 [MEDIUM] 188.83.72.48         External Automation
 [MEDIUM] 188.83.72.61         Anomalous External Users
 [MEDIUM] 188.83.72.64         Anomalous External Users
 [MEDIUM] 188.83.72.166        External Automation
 [MEDIUM] 188.83.72.174        Anomalous External Users
 [MEDIUM] 188.83.72.182        Anomalous External Users
```

```
[MEDIUM] 188.83.72.185        External Automation
[MEDIUM] 188.83.72.210        Anomalous External Users
[MEDIUM] 192.168.101.11       New Source IPs
[MEDIUM] 192.168.101.14       HTTPS Data Exfiltration
[MEDIUM] 192.168.101.26       HTTPS Data Exfiltration
[MEDIUM] 192.168.101.68       New Destination IPs / ASNs
[MEDIUM] 192.168.101.93       New Source IPs
[MEDIUM] 192.168.101.117      BotNet Beaconing
[MEDIUM] 192.168.101.138      New Destination IPs / ASNs
[MEDIUM] 192.168.101.148      DNS Anomalies
[MEDIUM] 192.168.101.157      BotNet Beaconing
[MEDIUM] 192.168.101.186      New Destination IPs / ASNs
[MEDIUM] 192.168.101.187      HTTPS Data Exfiltration
[MEDIUM] 192.168.101.188      BotNet Beaconing
[MEDIUM] 192.168.101.197      HTTPS Data Exfiltration
[MEDIUM] 192.168.101.208      HTTPS Data Exfiltration
```

False Negative Check

A false negative is a compromised device that the pipeline did not flag, the most dangerous failure mode in a detection system. To verify that no real anomaly was silently missed, we independently recomputed the highest metric value observed among all unflagged devices for each volume-based rule, and compared it against the threshold. If any unflagged device sits suspiciously close to the boundary, the threshold may be too conservative.

- For HTTPS upload volume, the highest unflagged device reached 105.2 MB against a threshold of 116.6 MB, a comfortable 11.4 MB gap.
- For DNS flow volume, the closest unflagged client produced 1,307 queries against a threshold of 1,399, leaving a 92-flow buffer. The two beaconing rules required closer inspection.
- For HTTPS beaconing, .160 sits at an interval standard deviation of 18.44 s, only 0.37 s above the 18.81 s floor, but its regularisation ratio is 1.17×, below the 1.5× guard: the device was already similarly regular during training, so the observed pattern is a stable baseline characteristic rather than a new anomaly.
- For DNS beaconing, the closest unflagged device has an interval standard deviation of 63.26 s (0.47 s above the 63.73 s floor) and a DNS flow count below the volume threshold; timing regularity without abnormal volume does not constitute meaningful beaconing.
- Finally, for the fan-out rule, .114 at 234 unique external IPs sits 24 IPs below the 258 threshold with no other anomalous signals, confirming the boundary is not cutting too close to normal behaviour.

Highest Confidence Detections

The 7 HIGH confidence devices are those flagged by two or more completely independent detection rules. Independence here is not just statistical, each rule operates on a different metric, a different protocol, and a different aspect of behaviour.

- The BotNet rule measures traffic timing regularity per protocol.
- The DNS rule measures query volume and destination.
- The HTTPS rule measures upload volume.
- The Geo rule measures destination countries.
- The New Destination rule measures new ASNs and new internal connections.
- The Fan-out rule measures the count of unique external destinations contacted.
- The New Source IP rule checks set membership against training

None of these share an input or a computation path. When two or more of them agree on the same device, they are making the same accusation from entirely different angles.

IP	Rules Triggered	Interpretation
192.168.101.41	DNS Volume + BotNet	39,493 DNS queries at an exact 0.05 s interval: extreme C2 beaconing via internal DNS relay, confirmed by both volume and timing
192.168.101.23	DNS Volume + BotNet	8,651 DNS queries at 0.05 s: same C2 pattern, smaller scale, same malware family
192.168.101.201	DNS Volume + BotNet	2,941 DNS queries at exact 0.05 s intervals confirmed independently by volume analysis (8.4σ above mean) and timing regularity: strong single-protocol C2 beaconing
192.168.101.207	HTTPS Exfiltration + Fan-out + DNS Anomalies	119 MB uploaded over HTTPS (3σ above threshold); 273 unique external IPs contacted (3.3σ above mean); 1,418 DNS queries at an exact 0.05 s beacon interval: three independent rules each confirming a different facet of the same compromise
192.168.101.72	New Geo + New Destination + Fan-out	Contacted 15 new countries including RU, IR, UA; 354 new-to-network ASNs pointing at Russian hosting infrastructure; 811 unique external IPs (14.9σ): geographic, infrastructure, and volume expansion confirmed by three independent rules

IP	Rules Triggered	Interpretation
192.168.101.36	New Geo + New Destination + Fan-out	904 flows to 17 new countries including RU, IR, IQ, KZ; 1,015 flows to 287 new-to-network ASNs; 639 unique external IPs (11.2σ): same three-rule as .72, consistent with coordinated exfiltration
192.168.101.125	New Geo + New Destination + Fan-out	523 flows to 11 new countries including RU, IR, UA; 598 flows to 169 new-to-network ASNs; 409 unique external IPs (6.2σ): third device in the coordinated group, confirmed by the same three independent rules

Attack Patterns

Grouping the 31 flagged devices by behaviour reveals nine distinct patterns active in the test period:

Pattern	Devices	Description
Mass HTTPS Exfiltration	.187, .14, .208	4.4–7.6 GB uploaded in a single day, 183–316σ above baseline: bulk data dumps, likely automated
Moderate HTTPS Exfiltration	.197, .26, .207	119–259 MB uploaded, 3–9σ above the 116.6 MB threshold
DNS C&C Beaconing	.41, .23, .201, .148	Exact 0.05 s query interval to internal resolvers: malware heartbeat routed through internal DNS relay
HTTPS C&C Beaconing	.117, .157, .188	Check-in intervals clustering between 1.00–1.03 s across three independent devices: single malware family, shared hardcoded timer
Exfiltration + C2 Beacon	.207	HTTPS exfiltration (119 MB, 3σ) running alongside a DNS C2 beacon (1,418 queries at 0.05 s intervals) and anomalous fan-out to 273 unique external IPs (3.3σ): three independent rules each confirming a different facet of the same compromise
Lateral Movement	.68, .138, .186	Internal-to-internal TCP:443 traffic to IPs never seen as destinations in training; symmetric byte ratios ($\sim$1:1) inconsistent with web browsing: peer-to-peer command relay or lateral file transfer between compromised hosts
Geo / Infrastructure Expansion	.36, .72, .125	Traffic to countries and ASNs never seen in training, plus anomalous fan-out to 409–811 unique external IPs: three independent rules (geo, ASN, volume) each confirming the same coordinated campaign

Pattern	Devices	Description
Rogue Device / New Implant	.11, .93	Devices with zero training presence generating 1,508 and 3,788 flows in the test period: stealthy profiles (14–38 MB HTTPS, 199–502 DNS queries) calibrated to stay below every volume threshold; caught exclusively by the binary set membership check
External Upload Anomaly	.61, .64, .174, .182, .210	External clients breaking the tight 8.50 ratio invariant: three uploading proportionally more than normal, two downloading more than normal
External Automation	.32, .46, .48, .166, .185	Inter-flow interval std of 0.51–2.03 s, against a training floor of 16.16 s : timing precision incompatible with human browsing; may represent bots, scanners, or legitimate automated consumers requiring analyst follow-up

---

SIEM Integration

Sending the Alert

Once all anomalous IPs are identified, each one is reported to the Wazuh manager by sending a UDP syslog message in the format "Alarm UEBA <ip>".

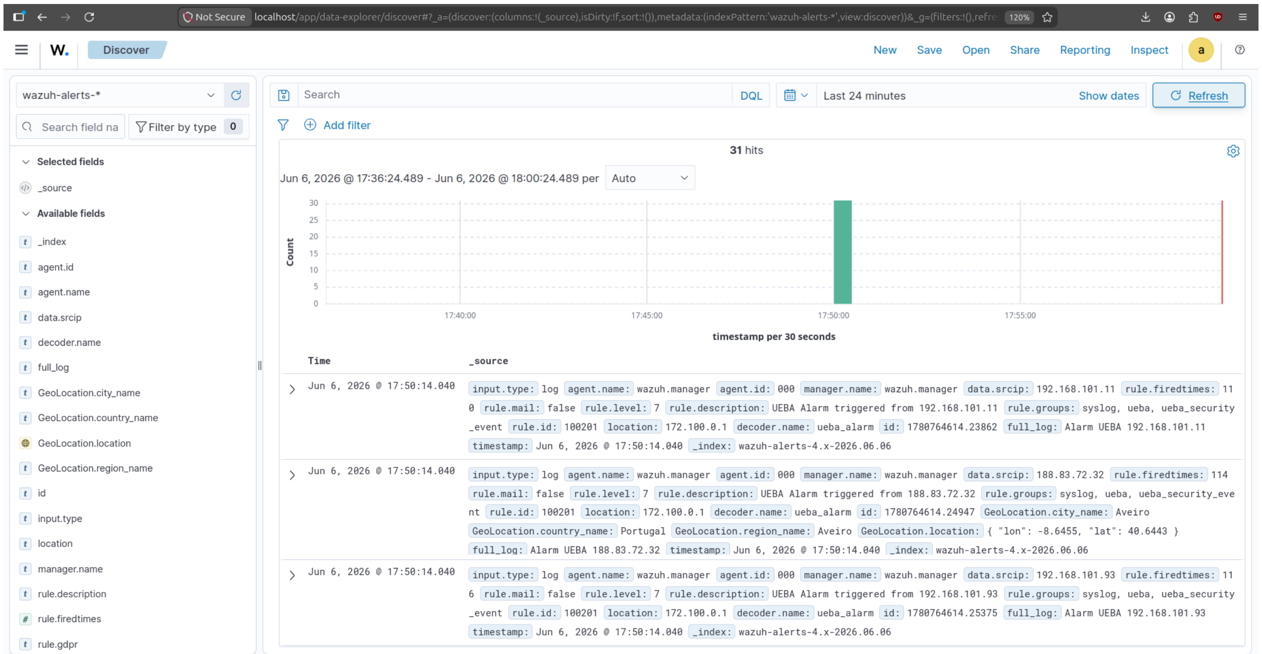
```
def send_syslog_alert(ip: str) -> None:
  with socket.socket(socket.AF_INET, socket.SOCK_DGRAM) as sock:
    sock.sendto(f"Alarm UEBA {ip}".encode(), (WAZUH_IP, WAZUH_PORT))
```

One packet is sent per flagged IP. All 31 were successfully delivered to 172.100.0.12:514:

```
[*] Sending UEBA alerts to Wazuh (172.100.0.12:514)...
      → Alarm UEBA 192.168.101.11
      → Alarm UEBA 192.168.101.93
      → Alarm UEBA 188.83.72.174
      ...
      → Alarm UEBA 192.168.101.188   (31 total)
```

Dashboard

With the Wazuh stack running and all 31 syslog packets delivered, navigating to Security Events and applying the DQL filter rule.id: 100201 (as it was used in the classes) shows one event per flagged IP: 31 hits, each carrying data.srcip with the anomalous device's address, rule.level = 7, and decoder.name = ueba_alarm.



Conclusion

The UEBA module identified 31 anomalous devices (21 internal and 10 external) across nine distinct behavioural patterns in the test period. Every threshold was derived exclusively from the training dataset, ensuring that no test-period information influenced the baselines and that each detection reflects a genuine deviation from established normal behaviour.

The pipeline demonstrates that a small set of complementary statistical rules, each measuring a different dimension of behaviour, can cover a broad threat surface without redundancy. Cumulative upload volume catches both bulk and low-and-slow exfiltration; timing regularity exposes botnet implants that leave no volumetric footprint; geographic and infrastructure novelty flags coordinated reconnaissance campaigns; and set-membership checks catch rogue devices that deliberately stay below every volume threshold.

The seven HIGH-confidence detections, each corroborated by two or more independent rules measuring entirely different signals, represent the strongest candidates for immediate escalation. The 24 MEDIUM-confidence detections provide a prioritised triage list where every alert carries an explicit justification tied to a specific observed value and its distance from the training baseline, giving an analyst the context needed to make an informed decision rather than a bare IP address.